

ZenVendorPulse — Infrastructure and Software Requirements

Zensar for Shell

1. Cloud Platform & Managed Services

Service	Tier	Role
Azure VM	D4as v6 or D8as v6	FastAPI + MAF backend
Azure Static Web Apps	Standard	React SPA (CDN)
Azure PostgreSQL Flexible Server	GP+HA (prod)	Datastore
Azure Key Vault	Standard	Secrets via Managed Identity
Azure Front Door + WAF	Standard	TLS, OWASP, origin-lock
Azure App Insights + Azure Monitor	Pay-as-ingested	OTel telemetry + audit mirror
Azure Container Registry (ACR)	Standard	Backend images
Microsoft Foundry / Azure OpenAI	Responses API	LLM + content safety
Microsoft Graph / Entra ID	—	Outlook/calendar/Teams · SSO/RBAC

2. Software

2.1 Runtimes & Package Managers

Software	Version
Python	3.11.x
Node.js	20 LTS (≥20.19) or 22 LTS (≥22.12) — required by Vite 8
npm / pip / Git	10.x / 23+ / 2.40+

2.2 Software packages

Package	Version	Purpose
fastapi	0.115.6	Web framework / REST API
uvicorn[standard]	0.32.1	ASGI server
openai	≥1.50.0	LLM SDK (Foundry Responses API)
azure-ai-projects / azure-identity	≥1.0.0 / ≥1.17.0	Foundry client / Entra auth
react / react-dom	19.2.4	UI

2.3 AI / Agent Stack

Component	Version / Detail
MAF SDK	1.x (≈1.8, Jun 2026) — pin exact
Microsoft Foundry	Responses API + content safety
Model	gpt-4.1 / gpt-4o (LLMProvider abstraction; Anthropic selectable)

2.4 Developer & CI Tooling

Tool	Version	Purpose
Claude Code	latest	AI pair-programmer / doc generation
Azure CLI (az)	2.60+	Azure / Foundry / Graph auth
Docker	24+	Container image builds
ruff / eslint	latest	Lint (CI)

3. License's & Subscriptions

#	License	Tier
LIC1	Microsoft 365 (E3/E5) — service mailbox	Per-user
LIC2	Microsoft Entra ID	Incl. in M365
LIC3	Azure subscription	Consumption
LIC4	Microsoft Foundry / Azure OpenAI (gpt-4.1/gpt-4o)	Pay-per-token
LIC5	Azure DevOps / GitHub Enterprise	SaaS
LIC6	Claude Code (Anthropic) / GitHub Copilot	Pro/Max/Team seat

4. Access & Identity

Ref	Item	Owner
I1–I2	App registrations + certificate credentials	Shell Entra ID
I3	Admin consent: Mail.Send, Mail.Read, Calendars.ReadWrite, OnlineMeetings.ReadWrite.All, User.Read.All	Shell Entra ID (global admin)
I5–I6	Entra role groups + Application Access Policy (mailbox-scoped)	Shell IAM / Exchange

5. Assumptions & Constraints

- Targets Shell production stack (MAF + Foundry + Azure + Outlook). POC Google components removed.
- Secrets come from Key Vault via Managed Identity in deployed environments; .env is dev-only.

6. Developer Workstation

Component	Minimum	Recommended
CPU	i5 11th Gen / Ryzen 5 5000 (4c/8t)	i7/i9 12th Gen+ / Ryzen 7–9 (8c+)
RAM	32 GB	32–64 GB
Storage	512 GB SSD, ≥100 GB free	1 TB NVMe